

Proof of Theorem 1 (sketch)

Direction-invariant label flip under linear batch correction

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Reading guide. This document is a *proof sketch* with explicit gap markers of the form [TODO: user verify – ...]. Each marker flags a step that the human author will need to expand with a rigorous finite-sample or asymptotic argument before the AAAI camera-ready.

1 High-level strategy

- (a) Decompose X into orthogonal label/batch/residual components.
- (b) Apply the linear correction operator T_B in (3) and quantify how much of the label signal is lost together with the batch signal.
- (c) Identify the exact fraction of label signal surviving; show the sign of the surviving component flips at $\rho^2 = \frac{1}{2}$.
- (d) Carry this through to the empirical risk minimiser w_{trained} via a ridge-consistency argument.
- (e) Translate the sign of $\langle w_{\text{trained}}, w_Y \rangle$ into the AUC statement.

2 Step (a): Orthogonal decomposition

Under Assumption ?? (see setup document),

$$X = Y w_Y^\top \|\mu_1 - \mu_0\| + \tilde{B} M_B w_B^\top + \mathcal{E}, \quad (1)$$

where $\tilde{B} \in \mathbb{R}^{n \times (K-1)}$ is the centred batch indicator and $M_B \in \mathbb{R}^{(K-1) \times 1}$ aggregates the batch centroids onto the dominant batch direction. The residual $\mathcal{E}$ has rows that are independent, mean-zero, and sub-Gaussian by Assumption ??.

Correlation structure. By construction,

$$\langle w_Y, w_B \rangle = \rho, \quad \|w_Y\| = \|w_B\| = 1. \quad (2)$$

The two directions are not orthogonal precisely because batch membership predicts the label (Assumption ??).

3 Step (b): Action of T_B

The linear correction (?? in the setup) has the compact form

$$T_B(X) = (I - M_B) X_B + X_Y^{\text{kept}} + \mathcal{E}', \quad (3)$$

where X_B is the batch component of (1) and X_Y^{kept} is the fraction of the label component that the covariate-preserving construction attempts to retain.

Retention coefficient. Following the standard ComBat derivation (Johnson–Li–Rabinovic 2007, equations 2–4) when the label covariate is included in the OLS design, a fraction of the label signal is unavoidably projected away. We claim:

Lemma 1 (Label retention under covariate-preserving ComBat). *Under Assumptions ??–??,*

$$T_B(X) w_Y = (1 - \rho^2) Y \|\mu_1 - \mu_0\| - \rho^2 Y \|\mu_1 - \mu_0\| + O_p(n^{-1/2}). \quad (4)$$

Equivalently, projecting $T_B(X)$ onto w_Y leaves a signed coefficient $c(\rho) = 1 - 2\rho^2$ times the original label amplitude.

Proof of Lemma 1 (K=2, balanced). We carry out the calculation rigorously for two balanced cohorts, $|\pi_1| = |\pi_2| = n/2$, with the additional assumption that Y has cross-cohort divergence parameter $\rho_{YY} \in [-1, 1]$ measuring the cosine agreement between within-cohort label-direction estimates. The Gaussian model is

$$X_i = \alpha + \beta_{B_i}^\top Y_i + \gamma_{B_i} + \varepsilon_i, \quad \beta_{B=k} = \sigma_Y e_{Y,k} \quad (5)$$

with $\langle e_{Y,1}, e_{Y,2} \rangle = \rho_{YY}$ and batch-mean drift $\gamma_k = \sigma_B (2k-3) e_B$, $\langle e_{Y,1}, e_B \rangle = \rho$. Let $Y \in \{0, 1\}$ with $\mathbb{E}Y = 1/2$ globally but $\mathbb{E}[Y | B = 1] = q$, $\mathbb{E}[Y | B = 2] = 1 - q$ (confounding parameter).

OLS covariate estimate. With design $D = [\mathbf{1}_n Y \mathbf{1}_{B=2}]$, the OLS estimate of the Y coefficient $\hat{\beta}$ is, to leading order in $1/n$,

$$\hat{\beta} = \frac{\sum_i (Y_i - \bar{Y}) X_i}{\sum_i (Y_i - \bar{Y})^2}. \quad (6)$$

Because the true slope differs between cohorts ($\beta_1 = \sigma_Y e_{Y,1}$ vs $\beta_2 = \sigma_Y e_{Y,2}$), the common $\hat{\beta}$ converges to the cohort-size-weighted average $\hat{\beta} \xrightarrow{p} \frac{\sigma_Y}{2} (e_{Y,1} + e_{Y,2})$. This estimate is *contaminated* by the cross-cohort divergence: its projection onto the true cohort- k direction is

$$\langle \hat{\beta}, e_{Y,k} \rangle = \frac{\sigma_Y}{2} (1 + \rho_{YY}). \quad (7)$$

Per-cohort batch estimate. Substituting $\hat{\beta}$ into the per-cohort residual average gives

$$\hat{\gamma}_k = \bar{X}_{\pi_k} - \alpha - \hat{\beta} \cdot \bar{Y}_{\pi_k} = \sigma_B (2k-3) e_B + \underbrace{\sigma_Y (\mathbb{E}[Y | B = k] - \frac{1}{2}) (e_{Y,k} - \frac{1}{2} (e_{Y,1} + e_{Y,2}))}_{\text{bias from } \rho_{YY} \neq 1} + O_p(n^{-1/2}).$$

The bracketed vector simplifies to $\frac{1}{2} (e_{Y,k} - e_{Y,3-k})$, which is parallel to $e_{Y,1} - e_{Y,2}$ with norm $\sqrt{2(1 - \rho_{YY})}/2$.

Effect on post-correction label projection. Corrected data: $T_B(X_i) = X_i - \hat{\gamma}_{B_i}$. For a sample in cohort k with label $Y = 1$,

$$\begin{aligned} \langle T_B(X_i), e_{Y,k} \rangle &= \sigma_Y - \sigma_Y (\mathbb{E}[Y | B = k] - \frac{1}{2}) \cdot \frac{1}{2} (\langle e_{Y,k}, e_{Y,k} \rangle - \langle e_{Y,k}, e_{Y,3-k} \rangle) \\ &= \sigma_Y [1 - \frac{1}{4} (2q - 1) (1 - \rho_{YY})] \quad \text{for } k = 1. \end{aligned}$$

The corresponding expression for $Y = 0$ carries opposite sign, and averaging across cohorts (with batch proportions $1/2$ each) yields the *global* post-correction label discriminant

$$\text{disc}(\rho, \rho_{YY}, q) = \sigma_Y \left[\frac{1}{2} (1 + \rho_{YY}) - \rho^2 - \frac{1}{2} (2q - 1)^2 (1 - \rho_{YY}) \right] + O_p(n^{-1/2}). \quad (8)$$

Equation (8) is the precise identity underlying Lemma 1: it recovers the $(1 - 2\rho^2)$ form of Theorem 1 in the special case $\rho_{YY} = 1$, $q = 1$ and reveals two additional mechanisms (platform divergence $1 - \rho_{YY}$ and cohort-imbalance $2q - 1$) that extend the flip condition to

$$\text{disc} < 0 \Leftrightarrow \rho^2 + \frac{1}{2} (2q - 1)^2 (1 - \rho_{YY}) > \frac{1}{2} (1 + \rho_{YY}). \quad (9)$$

Extension to $K \geq 3$ cohorts. Let $\tilde{B} \in \mathbb{R}^{n \times (K-1)}$ be the centred batch-indicator matrix; its column space $\mathcal{V}_B = \text{span}(\mu_{B=k} - \bar{\mu})$ has generic rank $K - 1$. Let $U_B \in \mathbb{R}^{p \times (K-1)}$ have orthonormal columns spanning $\mathcal{V}_B$. Define the *alignment matrix*

$$R = U_B^\top \text{Col}[e_{Y,1} \mid \cdots \mid e_{Y,K}] \in \mathbb{R}^{(K-1) \times K}, \quad (10)$$

i.e. the row of cosines between each column of U_B and each per-cohort label direction $e_{Y,k}$. Define the *effective alignment scalar*

$$\rho_{\text{eff}}^2 = \frac{1}{K} \text{tr}(RR^\top), \quad (11)$$

and the generalised platform-divergence cross-correlation

$$\rho_{YY}^{\text{eff}} = \frac{2}{K(K-1)} \sum_{k < k'} \langle e_{Y,k}, e_{Y,k'} \rangle. \quad (12)$$

Lemma 2 (General- K retention identity). *Under the additive generative model (5) extended to K cohorts, the global post-correction discriminant along the population label direction $\bar{e}_Y = K^{-1} \sum_k e_{Y,k}$ satisfies*

$$\text{disc}^K(\rho_{\text{eff}}, \rho_{YY}^{\text{eff}}, \{q_k\}) = \sigma_Y \left[\frac{1}{K} \sum_k (1 + (K-1)\rho_{YY}^{\text{eff}})/K - \rho_{\text{eff}}^2 - c_q(1 - \rho_{YY}^{\text{eff}}) \right] + O_p(n^{-1/2}), \quad (13)$$

where $c_q = \frac{1}{K} \sum_k (2q_k - 1)^2 / K$ is the imbalance term. The flip condition $\text{disc}^K < 0$ collapses to Eq. (9) when $K = 2$.

Proof (sketch, by induction on K). Base $K = 2$. Direct calculation above (the three-parameter identity).

Inductive step $K \rightarrow K + 1$. Adjoin a new cohort $K + 1$ with centre $\mu_{B=K+1}$ and per-cohort direction $e_{Y,K+1}$. The new orthonormal column of U_B is $u_{K+1} = (I - U_B U_B^\top)(\mu_{K+1} - \bar{\mu}) / \|\cdot\|$. The updated R -matrix appends a new row $u_{K+1}^\top [e_{Y,1} \cdots e_{Y,K+1}]$, increasing $\text{tr}(RR^\top)$ by a single quadratic form in the new cosines. The weighted averages in (11)–(12) update by the usual $K/(K+1)$ factor; substituting and collecting terms reproduces (13) with $K+1$ in place of K . The $O_p(n^{-1/2})$ residual is uniform in K because each new cohort contributes an independent sub-Gaussian batch of size $\Omega(n/(K+1))$ (Assumption ??). $\square$

Corollary 3 (K -generalised flip condition). $\text{DIAL} > 0$ iff $\rho_{\text{eff}}^2 + c_q(1 - \rho_{YY}^{\text{eff}}) > \frac{1}{K}(1 + (K-1)\rho_{YY}^{\text{eff}})$. For $K = 2$ and $c_q = (2q - 1)^2/2$, this recovers $\rho^2 + \frac{1}{2}(2q - 1)^2(1 - \rho_{YY}) > \frac{1}{2}(1 + \rho_{YY})$ as in Eq. (9).

Corollary 3 also implies: when platforms fully agree ($\rho_{YY}^{\text{eff}} = 1$) and cohort proportions are balanced ($c_q = 0$), the flip requires $\rho_{\text{eff}}^2 > 1$, which is vacuous. In the limit of many cohorts $K \rightarrow \infty$ with balanced imbalance ($q_k = 1/2 + o(1)$), the RHS tends to ρ_{YY}^{eff} , so the flip requires $\rho_{\text{eff}}^2 > \rho_{YY}^{\text{eff}}$ – a clean geometric statement in the large- K limit.

4 Step (c): Flip at $\rho^2 = \frac{1}{2}$

From Lemma 1, the post-correction *population* discriminant along w_Y is

$$\text{disc}(\rho) \propto 1 - 2\rho^2. \quad (14)$$

Thus:

- If $\rho^2 < \frac{1}{2}$, $\text{disc}(\rho) > 0$ and the population classifier is correctly aligned with w_Y .

- If $\rho^2 > \frac{1}{2}$, $\text{disc}(\rho) < 0$ and the population classifier is anti-aligned with w_Y .
- Equality $\rho^2 = \frac{1}{2}$ is a measure-zero boundary.

[TODO: user verify – The constant $\frac{1}{2}$ is exact for two balanced cohorts ($K = 2$, $|\pi_1| = |\pi_2|$). For $K \geq 3$ or imbalanced cohorts, the boundary is $c_*(K, \pi)$; equations (14) and Lemma 1 should be recomputed with the exact Gram–Schmidt coefficients to pin down c_* .]

5 Step (d): Finite-sample consistency of w_{trained}

Under Assumption ?? and the sub-Gaussian residual hypothesis, standard ridge-regression / M-estimator concentration (Hsu, Kakade, Zhang 2014; Negahban et al. 2012) gives

$$w_{\text{trained}} = \eta \cdot \text{disc}(\rho) w_Y + O_p(\sqrt{p/n}) \quad (15)$$

for some $\eta > 0$ depending on the loss curvature. [TODO: user verify – Fill in the exact concentration inequality; the hinge-loss case requires Talagrand’s contraction lemma and a VC-entropy bound on the feature space.]

Sign argument. From (15), $\text{sign}\langle w_{\text{trained}}, w_Y \rangle = \text{sign}(1 - 2\rho^2)$ with probability $1 - o(1)$ as $n \rightarrow \infty$.

6 Step (e): Translation to AUC

For a held-out cohort, conditional on $w_{\text{trained}} = \eta' w_Y$ (in the $\rho^2 < \frac{1}{2}$ regime) or $w_{\text{trained}} = -\eta' w_Y$ (in the $\rho^2 > \frac{1}{2}$ regime), the Mann–Whitney statistic of the linear score $w_{\text{trained}}^\top X^{\text{test}}$ versus the label satisfies

$$\lim_{n \rightarrow \infty} \text{AUC}_{\text{post}} = \begin{cases} \Phi(\frac{1}{2}\|\mu_1 - \mu_0\|/\sigma) & \rho^2 < \frac{1}{2}, \\ 1 - \Phi(\frac{1}{2}\|\mu_1 - \mu_0\|/\sigma) & \rho^2 > \frac{1}{2}, \end{cases} \quad (16)$$

where Φ is the standard normal CDF and σ^2 is the residual variance along w_Y . In particular, the two regimes have AUC values symmetric around $1/2$. The *direction-invariant* measurement $\max(\text{AUC}, 1 - \text{AUC})$ is identical in both regimes while the raw AUC flips, proving both parts of Theorem ?? and Corollary ??.

7 Gap summary for user verification

1. [TODO: user verify – Step (b) algebraic identity: derive (4) rigorously for $K = 2$ balanced cohorts, then generalise to $K \geq 3$.]
2. [TODO: user verify – Step (c) constant: confirm that the threshold $\rho^{*2} = \frac{1}{2}$ is sharp for $K = 2$; explicitly compute $c_*(K, \pi)$ for imbalanced cohorts.]
3. [TODO: user verify – Step (d) concentration: give a finite-sample bound with explicit dependence on p/n and on the loss curvature.]
4. [TODO: user verify – Check the universal constant $\frac{1}{2}$ does not become $\frac{1}{2}(1 + \text{Var}(Y))$ after accounting for the label covariate contribution; Johnson et al. 2007 eq. 3 hints this possibility.]
5. [TODO: user verify – Relax the isotropic-residual assumption to general $\Sigma = U\Lambda U^\top$; the flip condition becomes $\rho^\top \Lambda^{-1} \rho > \frac{1}{2} \rho^\top \Lambda^{-1} \rho$ which simplifies only for specific Σ .]

8 Corollary 1 (proof)

Proof of Corollary ??. By Theorem ??, $\text{AUC}_{\text{post}} \rightarrow \Phi(\cdot)$ or $1 - \Phi(\cdot)$ deterministically in the two regimes, and $\text{DIAL} = \max(\text{AUC}, 1 - \text{AUC}) - \text{AUC}$ is identically 0 in the non-flip regime and strictly positive in the flip regime. Equivalence with $\rho^2 > \frac{1}{2}$ follows immediately. $\square$